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Model characteristics

Input modality	Pre-trained ST FMs	Pretraining data	Scale	Architecture & characteristics	Params (M)	Venue
ST	Novae	Spatial transcriptomics	~30M cells	Graph attention (GATv2) over spatial neighborhoods	n.r.	Nat Methods 2025
	SToFM	Spatial transcriptomics	88M cells	Multi-scale Transformer for ST representation	~48*	ICML 2025
	AIDO.Tissue-60M	Multi-tissue ST + scRNA	~22M cells	Asymmetric enc–dec Transformer; generalist tissue foundation model	60	ICML 2025
	HEIST	Spatial transcriptomics	22.3M cells	Hierarchical graph Transformer, cell-type aware	~10*	ICLR 2026
	scGPT-spatial	ST profiles	30M profiles	scGPT Transformer backbone + MoE decoder	~55*	bioRxiv 2025
Pathology+ST	OmiCLIP	H&E patch–transcriptome pairs	2.2M pairs	CoCa visual–omics contrastive	~638*	Nat Methods 2025
	STPath	Whole-slide images	983 WSIs	Geometry-aware spatial Transformer, pathology-guided	n.r.	npj Digit Med 2025
scRNA+ST	Nicheformer	scRNA + ST	110M cells	Transformer, joint scRNA–ST pretraining	49.3	Nat Methods 2025
	OmniCell	scRNA + ST	~67M cells	Transformer, cell-centric cross-modal pretraining	~74	bioRxiv 2025

* Estimated from official config/checkpoint; n.r. = not reported.